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Combination of Multiple Chromophores Enhances Second-Harmonic Generation in Nonpolar Noncentrosymmetric Oxide: $CdTeMoO_6$

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Single crystals of $CdTeMoO_6$ have been grown from its stoichiometric melts. $CdTeMoO_6$ is non-hygroscopic and resistant to diluted HNO_3 aqua-solution. It is also thermally stable up to 766 ℃ and thereafter melts congruently. Transmittance spectra of $CdTeMoO_6$ show broad transmittance window in the range of 345 nm–5.4 μm. Based upon single-crystal X-ray diffraction analysis, $CdTeMoO_6$ is nonpolar noncentrosymmetric (NCS) and features a neutral layered structure consisting of three types of NCS chromophores, i.e. TeO_4 polyhedra with stereoactive lone pair, distorted MoO_4 tetrahedra, and regular CdO_4 tetrahedra, the former two being counter-packed and the last one being nonpolar. Such packing manner of the NCS chromophores is traditionally considered unfavorable to produce strong second-harmonic generation (SHG). However, powder SHG measurements reveal that $CdTeMoO_6$ displays remarkable SHG effect about 2 times that of polar $KTiOPO_4$ (KTP) at 1.064 μm. The first-principles electronic structure and optical properties calculations have confirmed the present of strong SHG effect in $CdTeMoO_6$, and have primarily revealed its origin.

1. Introduction

Nonlinear optical (NLO) materials are of topical interest and great importance owing to their uses in laser frequency conversion, optical parameter oscillators, and optical communication.[1, 2] The prerequisite of a NLO material is that it should be crystallographically noncentrosymmetric (NCS), and the common strategy to predesign NCS structure is to employ NCS chromophores as fundamental building units, including d^0 transition metal-centered polyhedron,[3] stereoactive lone-pair cation,[4-7] or borate π-orbital systems.[8-11] Note that the crystallographic NCS is not sufficient for excellent NLO performance of a material. In fact, many other merits are also indispensable, including chemically and thermally stable, broad transmittance window, easy-processing, and most importantly, strong NLO response.[12] In order to enhance NLO response of a material, it is generally accepted that the NCS chromophores should be packed in a "constructively" added manner. Note that this packing manner always gives rise to polar materials. In the past decades, a large amount of such materials have been discovered or predesigned,[13] such as stereoactive lone-pair oxides like $Cs_2I_4O_{11}$,[14] second-order Jahn-Teller (SOJT) distorted d^0 metal oxides like $A_3V_5O_{14}$ (A = K^+, Rb^+, or Tl^+),[3] π-orbital borates like $Na_2CsBe_6B_5O_{15}$,[15] $NaSr_3Be_3B_3O_9F_4$[16] and $K_3B_6O_{10}Cl$,[17] and many multi-chromophore oxides like $BaTeM_2O_9$ (M = Mo, W),[18] $TiO(IO_3)_2$[19], $AMoO_3(IO_3)$ (A = Rb, Cs),[20] $Li_2Ti(IO_3)_6$,[21] and $Pb_2B_5O_9I$,[22] and so on. In comparison, relatively few of nonpolar NCS oxides are NLO active, two well-known examples being KH_2PO_4 (KDP) and α-SiO_2.[23, 24]

The title compound was first synthesized in the latter half of the 1970s,[25, 26] and in 2001, its structure was determined by powder X-ray diffraction (PXRD) and electron diffraction analysis as belonging to the tetragonal space group of $P\bar{4}2_1m$.[27] To date, all studies on $CdTeMoO_6$ have been limited to the synthesis and catalysis properties of powders.[25-30] In this work, single crystals of $CdTeMoO_6$ have been successfully obtained, and its crystal structure has been accurately determined by single-crystal XRD analysis for the first time. Furthermore, chemical stability, thermal behavior, transmittance spectroscopy, and SHG properties of $CdTeMoO_6$ have been investigated in detail. Remarkably, although $CdTeMoO_6$ crystallizes in a nonpolar NCS space group, it exhibits a powder second-harmonic generation (SHG) efficiency of about 2 times that of polar KTP ($KTiOPO_4$). We note that this is the highest powder SHG effect among nonpolar oxides to date. In terms of nonpolar compounds, the containing NCS chromophores should be counter-packed so that the polarizations cancel out completely, or they are essentially nonpolar. However, as mentioned above, the counter-packing of NCS chromophores is traditionally considered unfavorable to produce strong SHG effect. Thus, it seems unusual that the nonpolar $CdTeMoO_6$ exhibits SHG effect even stronger than that of the highly polar material KTP. To further investigate its SHG behavior, we have done the first-principles electronic structure and optical properties calculations.

2. Experimental

2.1 Synthesis and crystal growth

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Polycrystalline samples of $CdTeMoO_6$ were synthesized by the traditional solid-state reaction techniques. The starting chemicals were TeO_2 (5N), MoO_3 (analytical grade), and CdO (analytical grade). A mixture of 32.0 TeO_2 (0.2 mol), 28.8 g MoO_3 (0.2 mol), and 25.7 g CdO (0.2 mol) was thoroughly ground, slowly heated to 550 ℃ at a rate of 20 ℃/h, and then sintered at this temperature for 72 h with several intermediate grindings. Polycrystalline $CdTeMoO_6$ powders were obtained and the phase purity was confirmed by PXRD analysis. The measured PXRD pattern matches the one calculated from single-crystal X-ray diffraction (XRD) analysis very well (Figure S1 in the Supporting Information). The differences in peak intensity for the same crystallographic index between the two patterns are believed to be caused by the preferential orientation of the powder samples.

Single crystals of $CdTeMoO_6$ were grown from its stoichiometric melts by a spontaneous nucleation approach. A platinum crucible containing $CdTeMoO_6$ polycrystalline powders was placed in the center of a vertical, programmable temperature furnace. The temperature was quickly increased to 850 ℃ and then held for 24 h to ensure the melt being homogenized. The melt was first cooled to 780 ℃ in an hour, then, cooled slowly at a rate of 1 ℃/h until plate-like crystals came forth on the matrix. Finally, the melt was rapidly cooled down to room temperature, and transparent $CdTeMoO_6$ crystals, as shown in Figure 1, were physically separated from the matrix.

2.2 Single-crystal X-ray diffraction

A $CdTeMoO_6$ crystal of $0.11 \times 0.05 \times 0.03$ mm^3 was selected for single-crystal XRD analysis. The diffraction data were collected with the use of graphite-monochromatized Mo *Kα* radiation (λ = 0.71073 Å) at 293(2) K on a Rigaku Mercury CCD/AFC diffractometer. The collection of the intensity data, cell refinement, and data reduction were carried out with the program Crystalclear.[31] The structure was solved by the direct method with program SHELXS and refined with the least-squares program SHELXL of the SHELXTL. PC suite of programs.[32] Final refinement includes anisotropic displacement parameters. Details of crystal parameters, data collection, and structure refinement are summarized in Table 1. The atomic coordinates and equivalent isotropic displacement parameters are listed in Table S1 in the Supporting Information. The selected bond distances and angles are presented in Table S2, and the anisotropic displacement parameters are listed in Table S3.

2.3 Chemical stability

$CdTeMoO_6$ single crystals with a weight of 0.158 g was soaked in distilled water, and the temperature was held at about 60 ℃ for 24 h. When the crystals were taken out, they are still transparent and their weight did not change. In order to further examine the chemical stability of $CdTeMoO_6$, the crystals were soaked in diluted HNO_3 solution (the volume ratio of concentrated HNO_3: H_2O = 1: 4) for 12 h at room temperature. The weight loss (0.02 g) is almost neglectable. These results demonstrate that the $CdTeMoO_6$ crystal is not only non-hygroscopic, but also resistant to diluted HNO_3 solution, showing excellent chemical stability.

2.4 Thermal stability

The thermal stability of $CdTeMoO_6$ was investigated by the differential scanning calorimetric (DSC) analysis on a NETZSCH STA 449C thermal analyzer (the DSC was calibrated with Al_2O_3). About 11.2 mg of $CdTeMoO_6$ were placed in a Al_2O_3 crucible, heated at a rate of 20 ℃/min from room temperature to 800 ℃, and then cooled to room temperature at the same rate. The measurement was carried out in an atmosphere of flowing N_2. The melted residues were examined and analyzed by PXRD after the experiment.

2.5 Transmittance spectroscopy

The UV and IR transmittance spectra were collected at room temperature on a PerkinElmer Lambda-35 UV/Vis/NIR Spectrometer from 250 to 1100 nm and a PerkinElmer Spectrum One FT-IR Spectrometer in the range of 1.28–10 μm, respectively. A $CdTeMoO_6$ crystal with a thickness of about 1 mm was used for the measurements without polishing.

2.6 Second-harmonic generation measurements

Powder SHG measurements were carried out by the Kurtz-Perry method.[33] Polycrystalline $CdTeMoO_6$ was ground and sieved into the following particle size ranges: 55–75, 75–103, 103–125, 125–150, 150–180, 180–212, and 212–300 μm. The samples were pressed between glass microscope cover slides and secured with tape in 1-mm thick aluminum holders containing an 8-mm diameter hole. They were then placed in a light-tight box and irradiated with a pulsed laser from a Q-switched Nd:YAG laser of wavelength 1064 nm. The intensity of the frequency-doubled output emitted from the samples was collected by a photomultiplier tube. Microcrystalline KTP samples with the same sizes served as the references.

2.7 Computational Details

The first-principles spin-polarized electronic structure calculations are performed using CASTEP, a plane wave pseudopotential total energy package based on density function theory (DFT).[34] The gradient corrected functionals developed by Perdew, Burke and Ernzerhof[35] is adopted to search the energy minimum of electronic structure and the potential energy of the interaction between ions and electrons is modeled by optimized norm-conserving pseudopotential[36] in kleinman-Bylander form,[37] with Cd $4d^{10}5s^2$, Te $5s^25p^4$, Mo $4d^55s^1$ and $O2s^22p^4$ treated as valence electrons. In order to consider the localized states in the transition elements Cd and Mo, the DFT+U method[38] with an additional on-site *d* orbital-dependent correlation Hubbard U[39] is adopted to calculate the electronic structure in $CdTeMoO_6$ as well. Kinetic energy cutoff 800eV and Monkhorst-Pack[40] k-point meshes with a density of $5 \times 5 \times 3$ points in the Brillouin zone are chosen. Our convergence tests show that the choice of above computational parameters is sufficiently accurate for this study.

It is well known that the band gap calculated by DFT is smaller than the experimental data due to the discontinuity of correlation function, so a scissor operator[41] is used to shift all the conduction band upward to agree with the measured data. Based on the calculated electronic band structure, we can obtain the imaginary part of the dielectric function from the electronic transition between valance band (VB) and conduction band (CB).[42] Consequently, the real part of the dielectric function, i.e., the refractive indices, can be determined by Kramers-Kronics transform.

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3. Results and discussion

3.1 Thermal behavior

As shown in Figure 2, the DSC data show a sharp endothermic peak around 766 ℃ in the heating curve and only one exothermic peak around 702 ℃ in the cooling curve, indicating that $CdTeMoO_6$ melts congruently at approximately 766 ℃. The melted residues exhibit a PXRD pattern in good agreement with that of the initial $CdTeMoO_6$ powders (Figure S2), further demonstrating that $CdTeMoO_6$ is a congruently melting compound. Therefore, large crystals of $CdTeMoO_6$ can be grown from its stoichiometric melts, which was confirmed by our crystal growth experiments. Obviously, the thermal stability of $CdTeMoO_6$ (congruently melting at 766 ℃) is better than other molybdenum tellurite materials such as $Na_2Te_3Mo_3O_{16}$ (incongruently melting at 468 ℃),[43] $Cs_2TeMo_3O_{12}$ (incongruently melting at 494.95 ℃),[44] $BaTeMo_2O_9$ (incongruently melting at 650 ℃),[18] and $MgTeMoO_6$ (incongruently melting at 682.35℃),[45] making the $CdTeMoO_6$ crystal relatively favorable in industrial production and applications.

3.2 Crystal structure

Based upon single-crystal XRD analysis, $CdTeMoO_6$ crystallizes in the tetragonal crystal system with a nonpolar NCS space group of $P\bar{4}2_1m$. In the crystal structure of $CdTeMoO_6$, there are three types of NCS chromophores, i.e. TeO_4 polyhedra with stereoactive lone pair, MoO_4 and CdO_4 tetrahedra. As shown in Figure S3, TeO_4 polyhedra and MoO_4 tetrahedra are alternately interconnected *via* corner-sharing to form infinite *zigzag* chains that are further linked by CdO_4 tetrahedra, thereby constructing a neutral layered structure.

Cd^{2+}, Te^{4+}, and Mo^{6+} cations occupy only one crystallographically independent site within the asymmetric unit, respectively. As shown in Figure 3, Cd^{2+} and Mo^{6+} cations are four-coordinated to form regular and distorted tetrahedra, respectively, while Te^{4+} cations are coordinated to four oxygen atoms in distorted "seesaw" configuration. As we know, both Mo^{6+} and Te^{4+} cations are susceptible to SOJT distortions, but a SOJT distortion is not observed in Mo^{6+} because of its tetrahedral coordination. Thus, similarly as in the case of $Zn_2(MoO_4)(TeO_3)$,[46] only the Te^{4+} cations undergo SOJT distortions, resulting in the formation of a stereoactive lone-pair that creates a locally asymmetric TeO_4 coordination environment. For $CdTeMoO_6$, the Cd–O, Te–O, and Mo–O bond distances are all in the normal ranges. Basically, our result based upon single-crystal XRD analysis is consistent with the structure based upon PXRD analysis.[27]

3.3 Dipole moments

In order to better understand the asymmetrical coordination environments, we did calculate the local dipole moments for the TeO_4, MoO_4 and CdO_4 polyhedra using a bond-valence approach proposed by Poeppelmeier *et al*.[47, 48] Detailed calculation results are given in an Excel document as Supporting Information. Table 2 summarizes the direction and magnitude of the dipole moments for the TeO_4, MoO_4, and CdO_6 polyhedra. With respect to the TeO_4 polyhedra, the lone pair is given a charge of -2 and localized 1.25 Å from the Te^{4+} cation. As a result, vector summation over the TeO_4 geometry gave a net dipole moment of 11.07 Debye which is obviously larger than the corresponding average value (8.57 Debye).[49] The direction of the dipole moment is just opposite to the stereoactive lone pair, somewhat surprisingly pointing to the less negative oxygen atoms owing to the intra-octahedral distortion (Figure 3). It is notable that, although the MoO_4 tetrahedra are not SOJT distorted, they still show considerable dipole moment of 7.42 Debye owing to their distorted tetrahedral configuration. The dipole moment vectors for TeO_4 and MoO_4 are directed towards their local *C*2 direction, respectively, and they are all paralleled to the *c* crystallographic axis. With CdO_4 tetrahedra, they are inherently NCS, but they are also nonpolar attributable to their regular tetrahedral configuration. Thus, it is reasonable that the magnitude of dipole moment turns out to be zero for CdO_4 tetrahedra (Table 2). Based upon the calculations, the NCS chromophores including TeO_4 and MoO_4 polyhedra are counter-packed themselves and CdO_4 tetrahedra are nonpolar, thereby resulting in non-polarity for $CdTeMoO_6$.

3.4 Transmittance spectroscopy

Figure 4 shows the UV and IR transmittance spectra of the $CdTeMoO_6$ crystal at room temperature. The UV absorption edge of $CdTeMoO_6$ is located at 345 nm, which is in agreement with the crystal colorlessness. Accordingly, the band gap of $CdTeMoO_6$ is as large as 3.59 eV, implying that $CdTeMoO_6$ may have a considerable laser damage threshold. As can be seen in Figure 4, the $CdTeMoO_6$ crystal is transparent up to 5.4 μm, covering the atmospheric transparency band II (3–5 μm).[50, 51] This indicates that the $CdTeMoO_6$ crystal may find applications in the mid-IR region as a NLO material. In the nature of things, transmittance measurements on single crystal give more accurate results than the powder diffuse reflectance method.

3.5 Second-harmonic generation

Because $CdTeMoO_6$ crystalizes in a NCS space group, it is expected to possess NLO properties. As shown in Figure 5, the SHG measurements on the sieved samples suggest type-I phase-matchable behavior at 1.064 μm for $CdTeMoO_6$. Remarkably, $CdTeMoO_6$ shows the highest powder SHG efficiency among nonpolar oxides to date (about 2 times that of KTP at 1.064 μm). In comparison, some selected examples are listed as follows: KDP (d_{36} = 0.39 pm/V),[23] $Se_2B_2O_7$ (2.2 × KDP),[52] $Na_3VO_2B_6O_{11}$ (～1 × KDP),[53] β-$KVO_2(IO_3)_2(H_2O)$ (very weak),[54] $Bi_2(IO_4)(IO_3)_3$ (5 × KDP),[55] and $MgTeMoO_6$ (1.5 × KTP).[45] Note that d_{eff} = 17.7 × 10^{-9} esu (7.44 pm/V) for KTP.[56]

3.6 Computational calculations

The band structure of $CdTeMoO_6$ (Figure 6) shows that this crystal is an indirect-band gap semiconductor and the calculated band gap is 2.91 eV, smaller than experimental result by 0.68 eV. The partial density of state (PDOS) is shown in Figure 7, from which some characters can be deduced: (1) O 2s orbital is strongly located between -19 and -14 eV, with very slight hybridization with Te 5s 5p and Mo 4d orbitals. (2) The middle part of valence band (-10 ~ -7eV) consists of O 2p, Cd 4d, and Te 5s, and the small hybridization of Cd 4d and O 2p indicates a relatively weak Cd-O bond. (3) The orbitals that occupy the top of valence band (-7 ~ 0eV) and the bottom of conduction band (2 ~ 8 eV) is O 2p, Mo 4d and Te 5p. The linear and nonlinear

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optical properties are directly related with the electron transition between the energy levels closed to the forbidden band. Therefore, TeO_4 polyhedra and MoO_4 tetrahedra contribute a lot to the SHG effect, while CdO_4 tetrahedra contribute less. Comparing the structures and compositions of $CdTeMoO_6$ and $MgTeMoO_6$, we find that larger SHG effect of $CdTeMoO_6$ may arise from the CdO_4 tetrahedra following the polarizability trend $Cd^{2+} > Mg^{2+}$. Further calculations with the DFT+U method demonstrate that for a broad range of Hubbard potential (U from 2 to 6 eV) the electronic states at the top of VB and bottom of CB have no qualitative modification.

$CdTeMoO_6$ belongs to class $\bar{4}2m$ and has two non-vanishing tensors of second-order susceptibility. Under the restriction of Kleinman's symmetry,[57] only one independent SHG tensor (d_{14}) remains. Our calculations gave a result of d_{14} = 11.7 pm/V (27.9 $\times 10^{-9}$ esu) at λ = 1.064 μm, which matches the experimental data, about 2 times that of KTP (d_{eff} = 17.7 $\times 10^{-9}$ esu).[56] We note that the d_{14} value for $CdTeMoO_6$ is even larger than that of the largest SHG coefficient (being d_{31} = 9.88 pm/V) for $BaTeMo_2O_9$,[58] in which the SOJT distorted MoO_6 and TeO_3 polyhedra are packed in a "constructively" added manner. In addition, the calculated linear refractive indices for $CdTeMoO_6$ (Figure S4) reveal that the crystal has a large birefringence, i.e., Δn = 0.17 at λ = 1064 nm. The relatively large Δn value at λ = 1.064 μm is consistent with the experimentally phase-matchable behavior and the layered crystal structure. The DFT+U calculations show that the optical properties of $CdTeMoO_6$ are almost independent of the choice of U.

4. Conclusions

Single crystals of $CdTeMoO_6$ have been successfully grown from its stoichiometric melts and characterized by single-crystal XRD for the first time. $CdTeMoO_6$ is nonpolar NCS and constructed by three types of NCS chromophores, i.e. TeO_4 polyhedra with stereoactive lone pair, distorted MoO_4 tetrahedra, and regular CdO_4 tetrahedra, the former two themselves being counter-packed and the last one being nonpolar. Traditionally, such packing manner of NCS chromophores is considered unfavorable to generate strong SHG effect. However, $CdTeMoO_6$ exhibits remarkable powder SHG effect about 2 times that of polar KTP at 1.064 μm, which is in agreement with the calculated SHG coefficient. Based upon primary theory calculations, the unusual SHG effect is directly attributable to the three types of NCS chromopores. Consequently, we considered that the combination of multiple chromophores would enhance the SHG effect to the greatest extent even in the case of nonpolar NCS materials. We believe this will definitely shed useful insight on the further exploration of new NLO materials. Moreover, $CdTeMoO_6$ features type-I phase-matchable behavior, excellent chemical stability, high thermal stability with congruent-melting behavior, and broad transmittance window covering the atmospheric transparency band II, indicating that it is a potential candidate for mid-IR NLO applications. Further researches are ongoing to study the detailed SHG mechanism.

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† Electronic Supplementary Information (ESI) available: [X-ray crystallographic file in CIF format for the $CdTeMoO_6$ single crystal, dipole moment calculation details in Excel format, experimental and calculated PXRD patterns for $CdTeMoO_6$, PXRD patterns of $CdTeMoO_6$ powders before and after melting, calculated linear refractive indices for $CdTeMoO_6$, atomic coordinates and equivalent isotropic displacement parameters (Å^2) for $CdTeMoO_6$, selected bond distances (Å) and angles (deg.) for $CdTeMoO_6$, anisotropic displacement parameters (Å^2) for $CdTeMoO_6$, CCDC 916482.]. See DOI: 10.1039/b000000x/
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The authors declare no competing financial interest.

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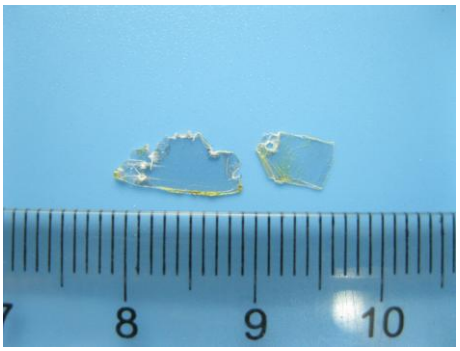

Fig. 1 Photograph of $CdTeMoO_6$ crystals.

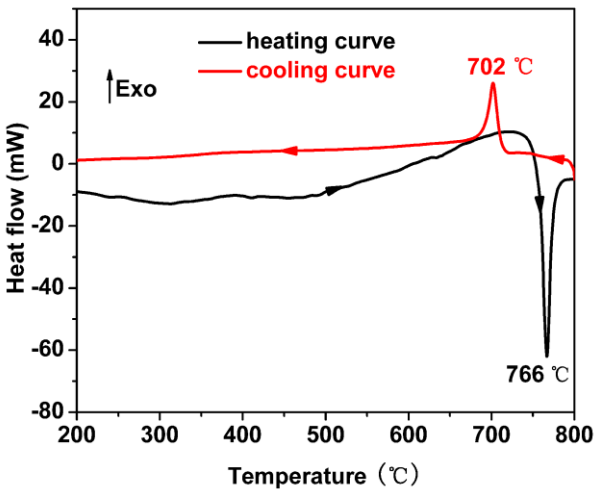

Fig. 2 DSC curves of $CdTeMoO_6$.

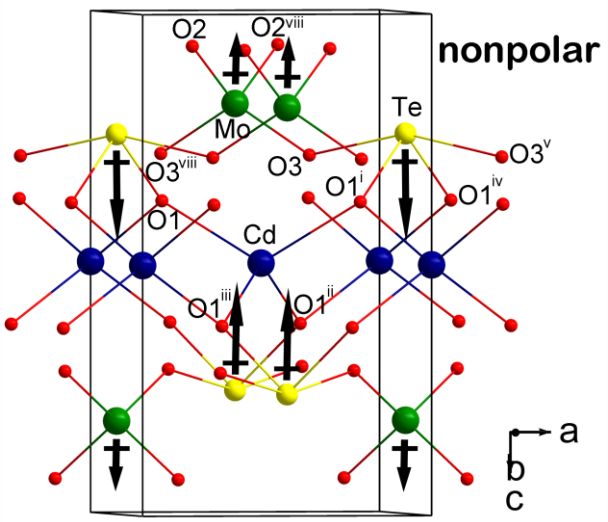

Fig. 3 The asymmetrical coordination environments for $CdTeMoO_6$. The arrows indicate the approximate direction of the dipole moments. Symmetry codes: (i) 1-x, 1-y, z; (ii) 1-y, x, 1-z; (iii) y, 1-x, 1-z; (iv) 1+x, y, z; (v) 2-x, 1-y, z; (viii) 1-x, -y, z.

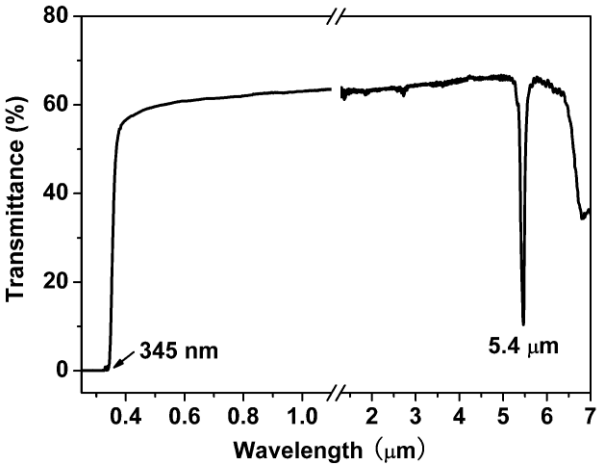

Fig. 4 UV and IR transmittance spectra of $CdTeMoO_6$.

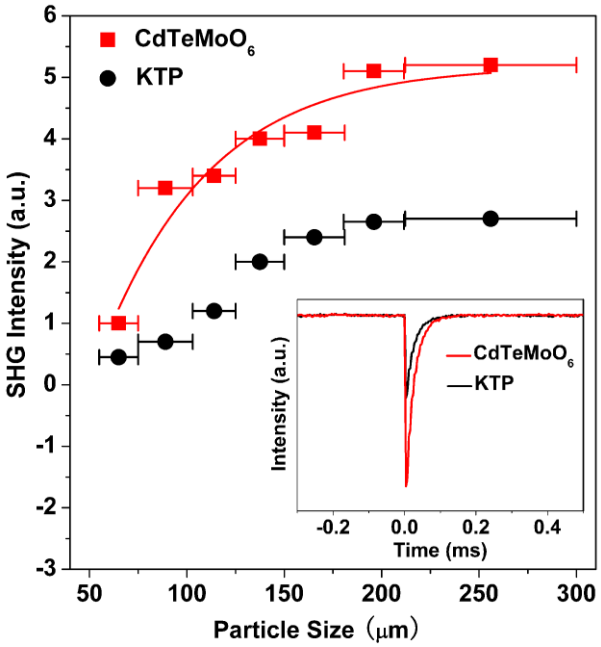

Fig. 5 Phase-matching data for $CdTeMoO_6$; and the inset is comparative SHG signals for the powder samples with same particle sizes, KTP (black) and $CdTeMoO_6$ (red).

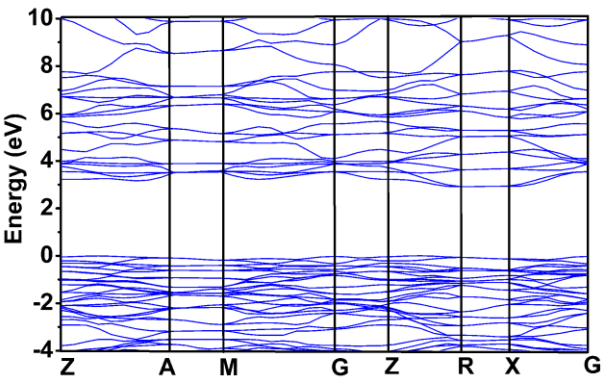

Fig. 6 The band structure of $CdTeMoO_6$.

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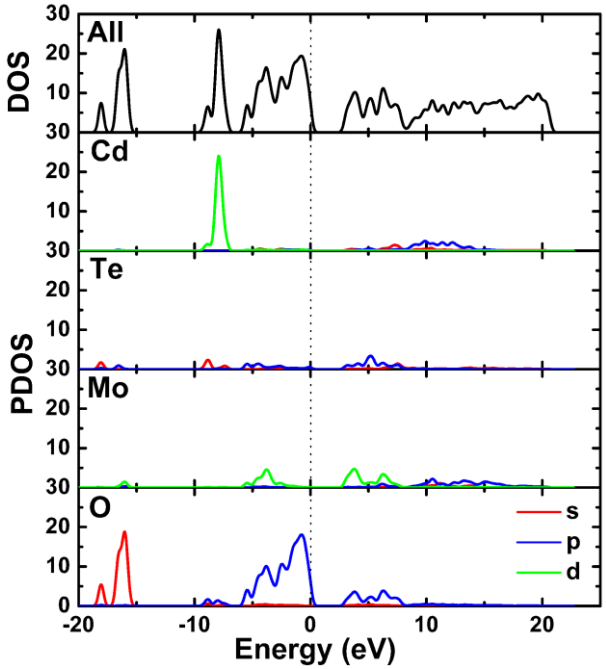

Fig. 7 Total and partial density of states of $CdTeMoO_6$.

Table 1 Crystal data and structure refinement for $CdTeMoO_6$.

Formula sum	$CdTeMoO_6$
Formula weight (g/mol)	431.95
Crystal system	tetragonal
Space group	$P\bar{4}2_1m$ (113)
Unit cell dimensions/Å	a = 5.2860(7), c = 9.0660(18)
V/Å^3	253.32(7)
Z	2
μ/mm^{-1}	12.265
F(000)	380
Data/restraints/parameters	338/0/29
R(int)	0.0316
GOF(F^2)	1.034
Flack parameter	0.01(6)
Final R indices [$F_o^2 > 2\sigma(F_c^2)$.][a]	R_1 = 0.0147, wR_2 = 0.0348
Final R indices (all data)[a]	R_1 = 0.0156, wR_2 = 0.0354

[a] $R_1 = \sum_2 \left\| F_o \right| - \left| F_c \right\| / \sum \left| F_o \right|$ and

$wR_2 = \left[\sum w(F_o^2 - F_c^2) / \sum wF_o^4 \right]^{1/2}$ for $F_o^2 > 2\sigma(F_c^2)$.

Table 2 The direction and magnitude for the dipole moments of the TeO_4, MoO_4, and CdO_4 polyhedra in $CdTeMoO_6$..

Species	Direction			Magnitude		Distortion symmetry
	x/a	y/b	z/c	Debye	esu cm Å^{-3}	
TeO_4	0	0	11.07	11.07	0.0875	$C2$
MoO_4	0	0	-7.42	7.42	0.0586	$C2$
CdO_4	0	0	0	0	0	-
Asymmetric unit	0	0	3.65	3.65	0.0289	-
Unit cell	0	0	0	0	0	-

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